Project Report — Day 1

Amazon Review NLP & Generative AI Web Application

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1. Project Objective

The objective of this project is to develop an interactive web application that analyzes Amazon e-commerce product reviews using Natural Language Processing (NLP) techniques. The system will gradually evolve from basic data analysis to a conversational AI system powered by Retrieval-Augmented Generation (RAG).

The final application aims to:

- Understand customer sentiment
- Extract useful product insights
- Provide summarized information
- Answer user queries using real reviews

Day-1 focuses on environment setup, dataset exploration, and building an initial Streamlit dashboard.

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2. Dataset Description

The dataset used is the **Amazon Tech Product Reviews dataset** obtained from Kaggle. It contains real customer reviews for electronic products sold on Amazon.

Main Features

- reviewText — Detailed customer review
- overall — Product rating (1–5 stars)
- summary — Short review title
- asin — Product identifier
- reviewerID — Reviewer identifier
- reviewTime — Date of review

This dataset is suitable for sentiment analysis, text mining, and generative AI applications.

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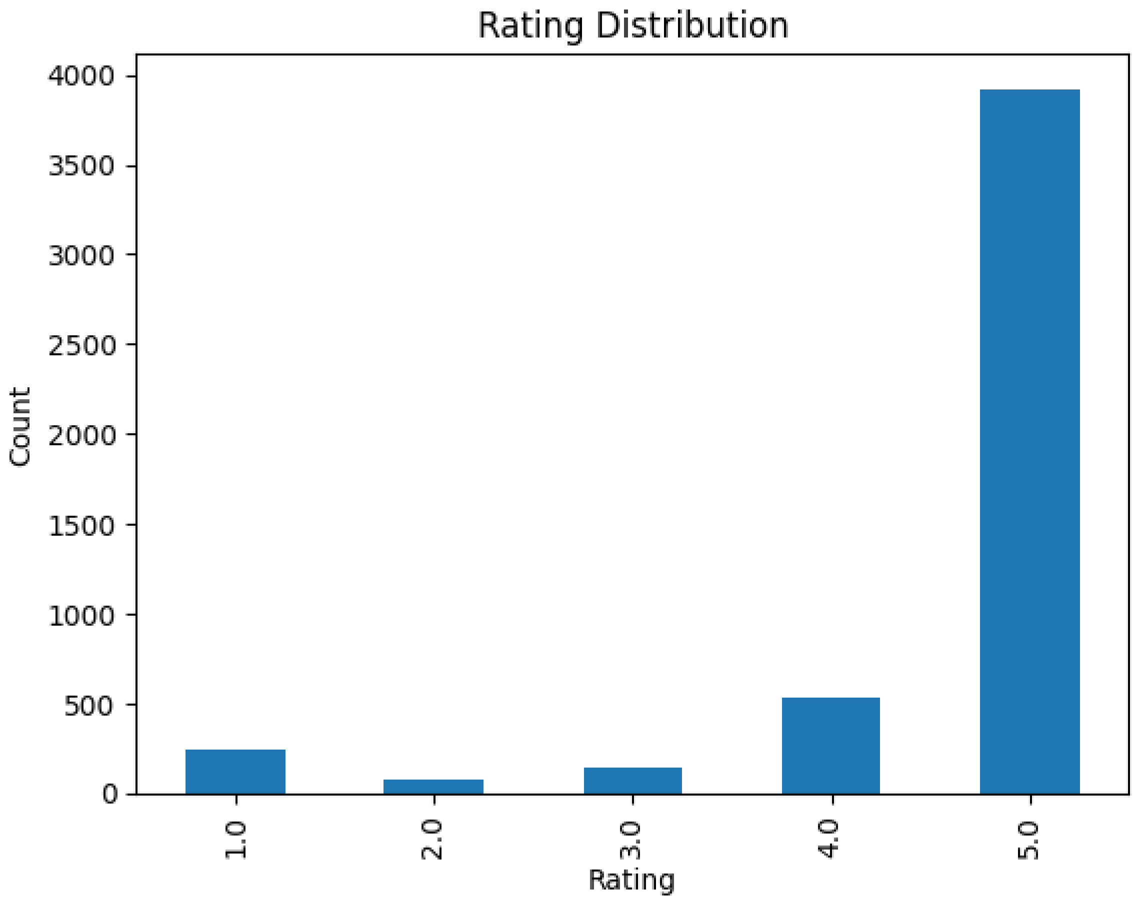
3. Exploratory Data Analysis (EDA)

3.1 Rating Distribution

Analysis of rating values shows that most reviews are rated 4 or 5 stars.
This indicates a strong positive bias in customer feedback and introduces class imbalance, which may affect machine learning models in later stages.



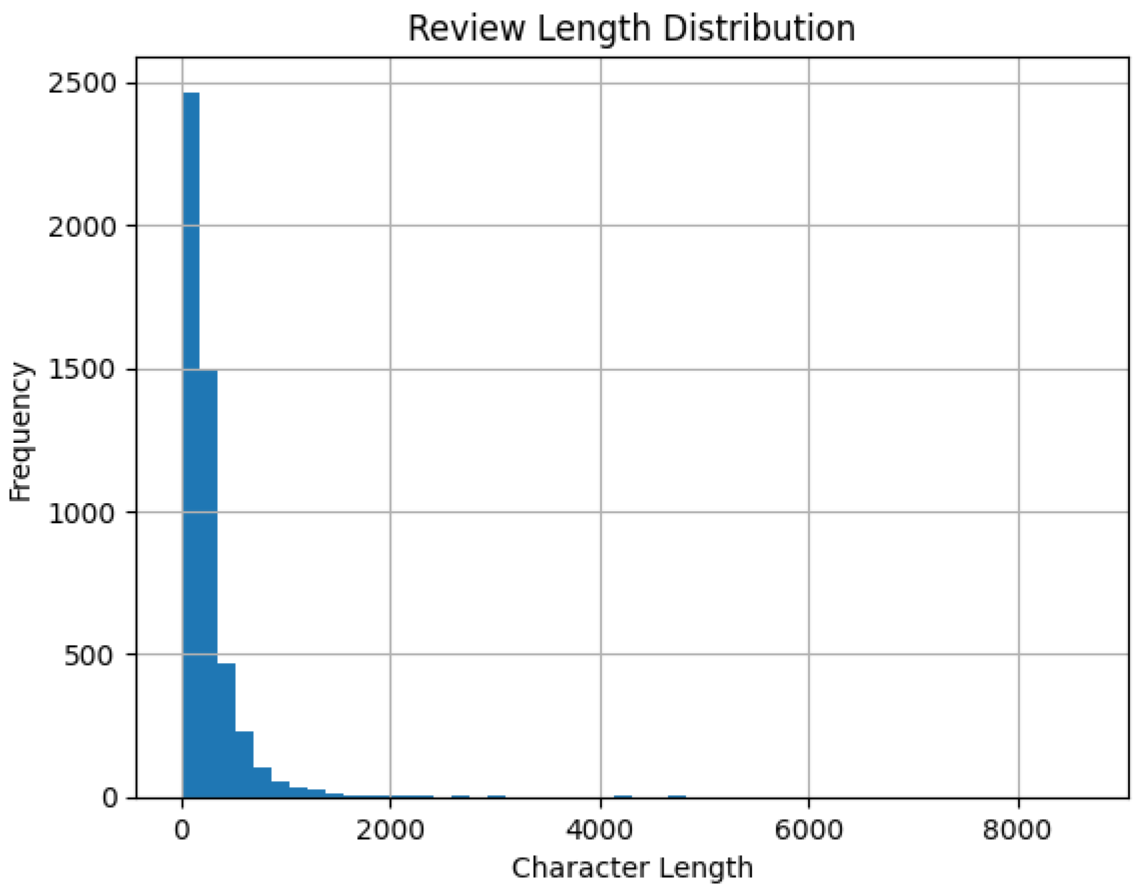
3.2 Review Length Analysis

A new feature review_length was created to measure the number of characters in each review.

Observations:

- Review sizes vary significantly
- Some users provide short feedback such as quick opinions
- Others provide detailed experiences and product evaluations

This variation suggests that future NLP processing must handle both short and long text inputs effectively.



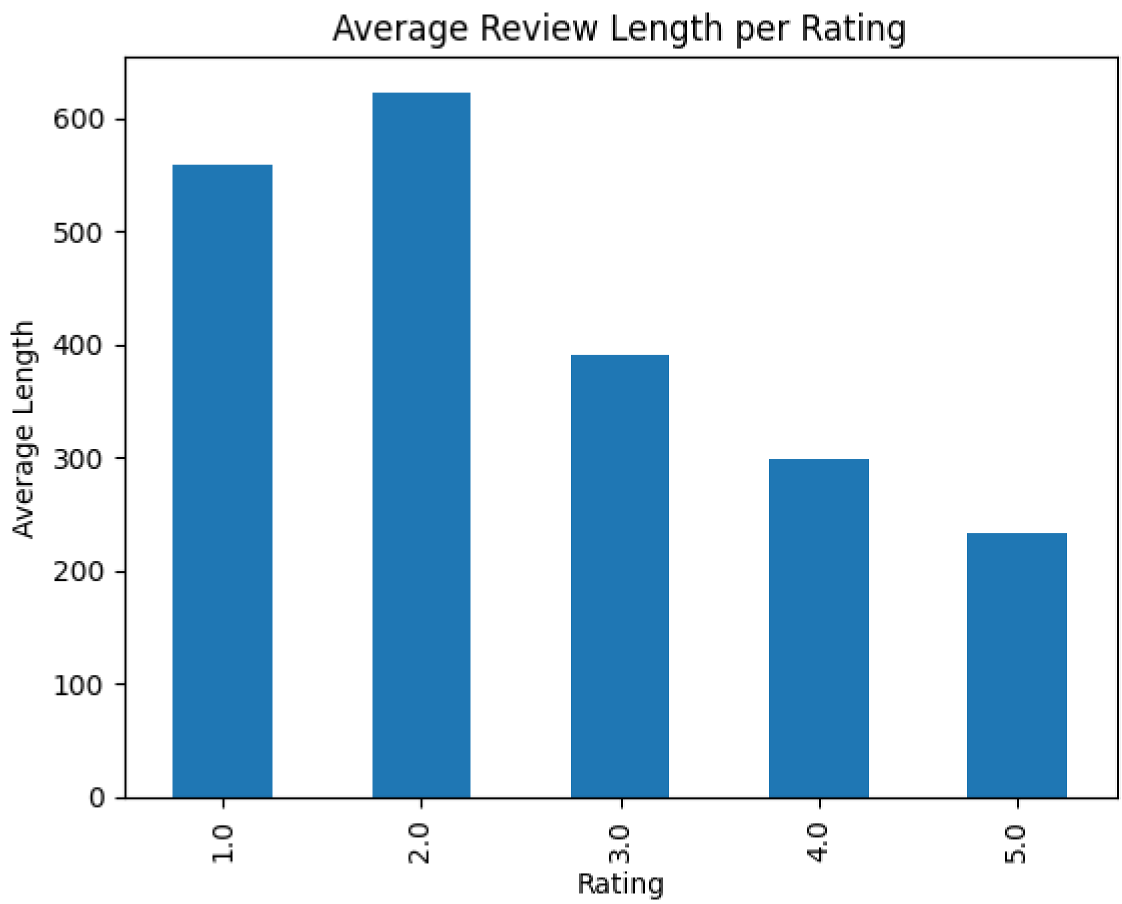
3.3 Rating vs Review Length

The relationship between rating and review length was analyzed.

Key findings:

- Low ratings often have longer reviews because customers explain issues in detail
- High ratings sometimes contain short appreciation comments
- Emotional intensity appears to influence how much users write

This insight will help in feature engineering and also guide chunk size selection when building the future RAG retrieval system.



3.4 Handling Missing Values

Some entries in the reviewText column contained missing values.
They were handled using:

fillna("")

This ensures stable processing and prevents runtime errors during text analysis.

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4. Streamlit Dashboard

A basic Streamlit application was developed to visualize and interact with the dataset.
The dashboard includes:

- Dataset preview table
- Rating distribution chart
- Review length distribution
- Rating vs review length comparison

This serves as the foundation for adding sentiment prediction and AI-based features in later stages.

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5. Conclusion

Day-1 successfully established the project environment and performed initial analysis of the dataset. Important behavioral patterns in customer reviews were identified, and a working interactive dashboard was created.

The next phase of the project will focus on text preprocessing and sentiment classification.